ಆದಿಕವಿ ಮಹರ್ಷಿ ವಾಲ್ಮೀಕಿ ವಿಶ್ವವಿದ್ಯಾಲಯ ರಾಯಚೂರು

ADIKAVI MAHARSHI VALMIKI UNIVERSITY RAICHUR

PROJECT REPORT

ON

"BRAIN STROKE DETECTION USING DEEP LEARNING"

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled **“Brain Stroke Detection Using Deep Learning”** has been successfully submitted by **Md Ather Ali** RegNo. **U23IA22S0077.**

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor’s degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

2.

Guide
Name & Guide signature

HOD
Department of Computer Applications

ACKNOWLEDGEMENT

I express my sincere gratitude to my guide**, Shri. Manjunath. V**, **Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

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I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

Date:- **MD ATHER ALI**

Regno:- U23IA22S0077

CANDIDATE DECLARATION

Md Ather Ali a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0077** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled **"Brain Stroke Detection Using Deep Learning"** submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of bachelor's degree in computer applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be slowly responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- **MD ATHER ALI**

Regno:- **U23IA22S0077**

Abstract

Brain tumour detection is a crucial task in medical imaging, as early diagnosis significantly improves patient survival rates. This project aims to develop an automated brain tumour detection system using deep learning techniques, integrating Convolutional Neural Networks (CNNs) for efficient feature extraction and classification. The system is trained on a medical imaging dataset containing MRI scans of brain tumours, categorized into different tumour types such as glioma, meningioma, and pituitary tumours, as well as non-tumour cases.

The model employs a pre-trained CNN architecture, such as VGG16, ResNet-50, or EfficientNet, for extracting spatial features from MRI scans. To enhance accuracy, transfer learning is utilized, allowing the model to leverage prior knowledge from large-scale image datasets. Additionally, data augmentation techniques, including rotation, scaling, and flipping, are applied to increase generalization and improve robustness against variations in medical images.

Early detection of brain stroke is critical for improving patient outcomes and guiding appropriate treatment strategies. This project leverages deep learning techniques to automate stroke detection, reducing the reliance on manual diagnosis and minimizing human error. The developed CNN-based model efficiently processes MRI or CT scan images, classifies them as stroke or non-stroke, and highlights the affected regions using thresholding techniques. By integrating an interactive GUI for image selection and real-time analysis, the system provides a user-friendly solution for medical practitioners and researchers. The implementation of transfer learning and advanced optimization techniques further enhances the model’s predictive accuracy, making it a reliable tool for preliminary brain stroke screening.

The classification process is optimized using fully connected layers, batch normalization, and dropout regularization to prevent overfitting. The model’s performance is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and AUC-ROC. The results demonstrate the effectiveness of deep learning in automating stoke detection, reducing diagnostic errors, and assisting radiologists in clinical decision-making.

This project has significant applications in medical diagnostics, enabling faster and more reliable detection of brain stroke, thus supporting early intervention and improved treatment planning. The integration of AI-driven solutions in healthcare contributes to the advancement of medical imaging technologies, enhancing the efficiency and accuracy of radiological assessments.

Keywords:

CNN, MRI, Brain Stroke, Deep Learning, ResNet-50, VGG16, EfficientNet, Transfer Learning, Medical Imaging, AUC-RO.

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